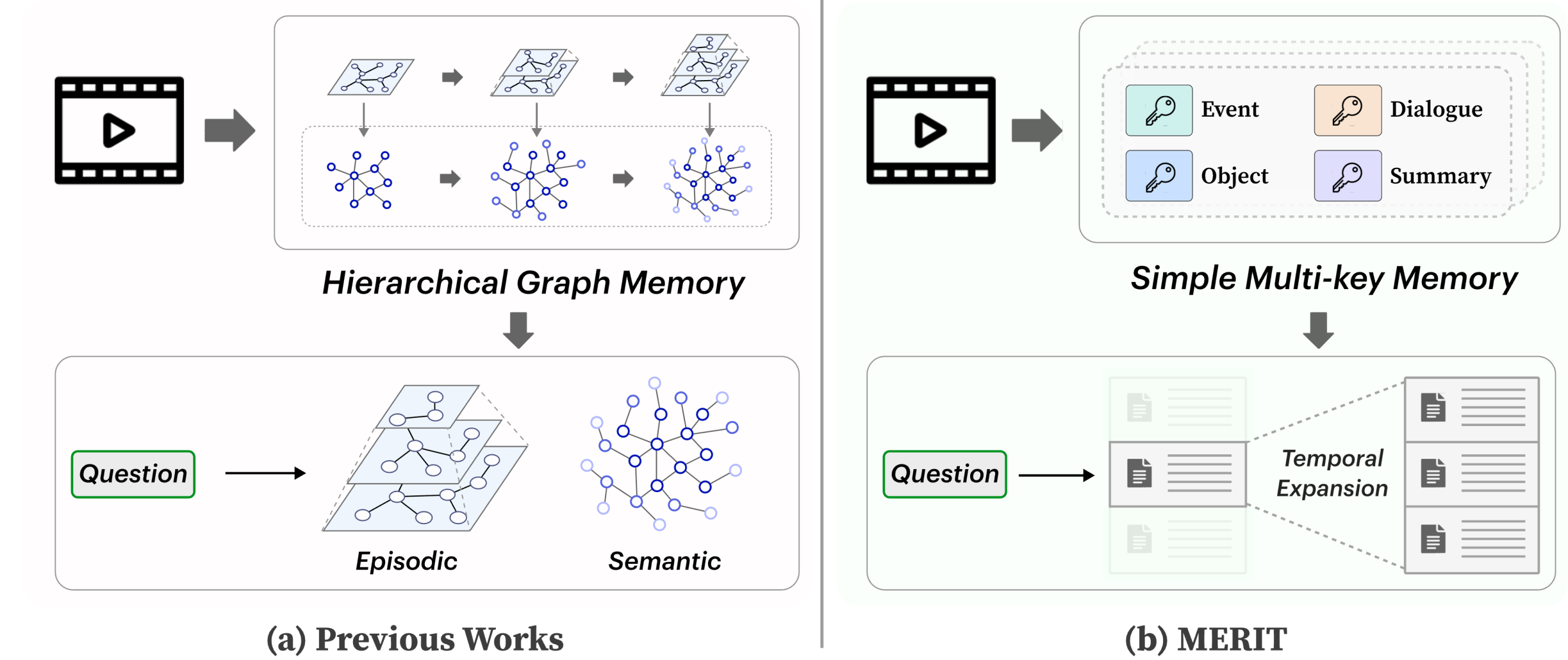


# Keep It Simple: Multi-Key Episodic Memory Retrieval for Ultra-Long Video Understanding

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## (1) Task: Ultra-Long Video QA

How to Handle Ultra-Long Video? – Memory Construction + Retrieval Inference



- **Prior work** pre-builds hierarchies and knowledge graphs **before the query is known**, thousands of propriety MLLM calls spent on an abstraction that *may not match what is finally asked*.
- **MERIT** keeps the memory **simple and high-fidelity**, retrieves with **high recall**, and defers **query-specific reasoning** to inference time on-demand.

### Motivation

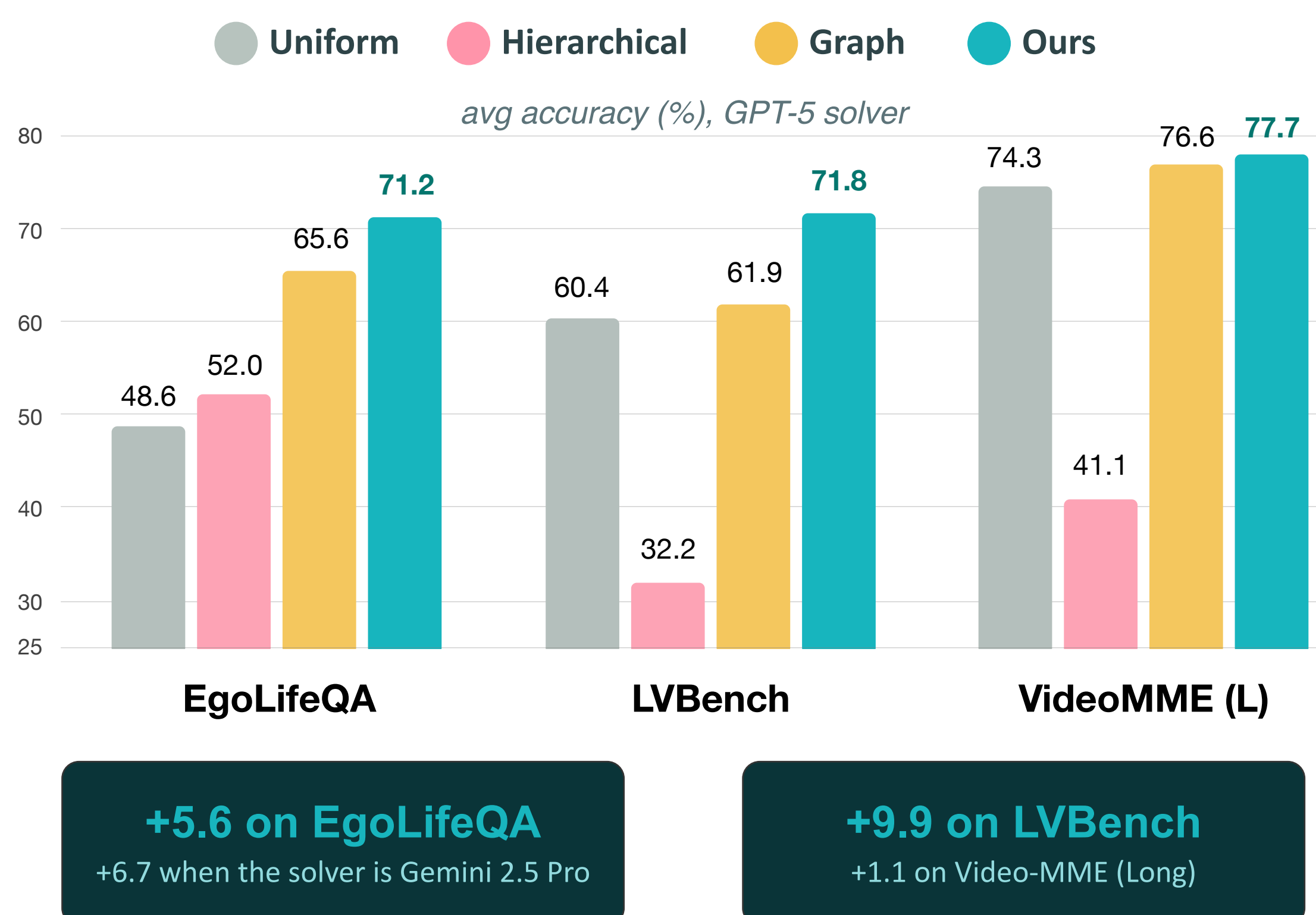
#### ① Retrieval is the Bottleneck

Ground Truth 30s Clip **82.4** >> Prev. SOTA **63.2**  
*About 20% of retrieval headroom*

#### ② Intelligence Allocation

*"Why spend your strongest model before you know the question?"*

## (4) Benchmark Results



### Ultra-long Video QA Results

| Model                            | EntityLog   | EventRecall | HabitInsight | RelationMap | TaskMaster  | Avg         |
|----------------------------------|-------------|-------------|--------------|-------------|-------------|-------------|
| <b>MLLMs (Uniform Sampling)</b>  |             |             |              |             |             |             |
| Qwen3-VL-8B [1]                  | 35.2        | 30.2        | 39.3         | 46.4        | 46.0        | 38.6        |
| Gemini 2.5 Pro [3]               | 43.2        | 40.5        | 41.0         | 55.2        | 52.4        | 46.4        |
| GPT-5 [12]                       | 47.2        | 42.1        | 47.5         | 53.6        | 55.6        | 48.6        |
| <b>Hierarchical Memory Based</b> |             |             |              |             |             |             |
| EgoRAG [20] (GPT-5)              | 40.0        | 56.3        | 62.3         | 54.4        | 52.4        | 52.0        |
| Ego-R1 [21] (3B)                 | 51.2        | 53.2        | 63.9         | 50.4        | 50.8        | 53.0        |
| <b>Graph Memory Based</b>        |             |             |              |             |             |             |
| LightRAG [1] (GPT-5)             | 40.8        | 48.4        | 67.2         | 50.4        | 44.4        | 48.8        |
| HippoRAG [9] (GPT-5)             | 48.8        | 60.3        | 70.5         | 60.8        | 66.7        | 59.6        |
| VideoRAG [11] (GPT-5)            | 49.6        | 56.3        | 67.2         | 55.2        | 54.0        | 55.4        |
| HippoMM [10] (GPT-5)             | 45.6        | 53.2        | 70.5         | 55.2        | 58.7        | 54.6        |
| M3-Agent [10] (7B)               | 44.4        | 54.8        | 62.3         | 56.8        | 54.0        | 53.5        |
| EGAgent [13] (Gemini 2.5 Pro)    | 54.4        | 57.1        | 60.3         | 62.4        | <b>74.6</b> | 57.5        |
| WorldMM [22] (Qwen3-VL-8B)       | 49.6        | 56.4        | 63.9         | 58.4        | 58.7        | 56.4        |
| WorldMM [22] (GPT-5)             | 62.4        | 64.3        | 75.4         | 62.4        | 71.4        | 65.6        |
| <b>Ours</b>                      |             |             |              |             |             |             |
| MERIT (Qwen3-VL-8B)              | 43.2        | 54.0        | 67.2         | 53.6        | 65.1        | 54.2        |
| MERIT (Gemini 2.5 Pro)           | 60.8        | 61.1        | 65.6         | 65.6        | 73.0        | 64.2        |
| MERIT (GPT-5)                    | <b>67.2</b> | <b>70.6</b> | <b>73.8</b>  | <b>74.4</b> | 71.4        | <b>71.2</b> |

### Hour-long Video QA Results

| Model                            | LVBench     | Video-MME(L) |
|----------------------------------|-------------|--------------|
| <b>MLLM Uniform Sampling</b>     |             |              |
| Qwen3-VL-8B [1]                  | 48.3        | 61.0         |
| Gemini 2.5 Pro [3]               | 57.0        | 55.7         |
| GPT-5 [12]                       | 60.4        | 74.3         |
| <b>Hierarchical Memory Based</b> |             |              |
| EgoRAG [20] (GPT-5)              | 32.2        | 41.1         |
| Ego-R1 [21] (3B)                 | 34.1        | 42.7         |
| <b>Graph Memory Based</b>        |             |              |
| LightRAG [1] (GPT-5)             | 30.4        | 46.6         |
| HippoRAG [9] (GPT-5)             | 54.0        | 52.1         |
| VideoRAG [11] (GPT-5)            | 33.1        | 55.4         |
| HippoMM [10] (GPT-5)             | 38.2        | 41.6         |
| M3-Agent [10] (7B)               | 49.3        | 55.3         |
| EGAgent [13] (Gemini 2.5 Pro)    | -           | 74.1         |
| WorldMM [22] (GPT-5)             | 61.9        | 76.6         |
| <b>Ours</b>                      |             |              |
| MERIT (Gemini 2.5 Pro)           | -           | 76.3         |
| MERIT (GPT-5)                    | <b>71.8</b> | <b>77.7</b>  |

## (6) Neighbor Filtering Effect

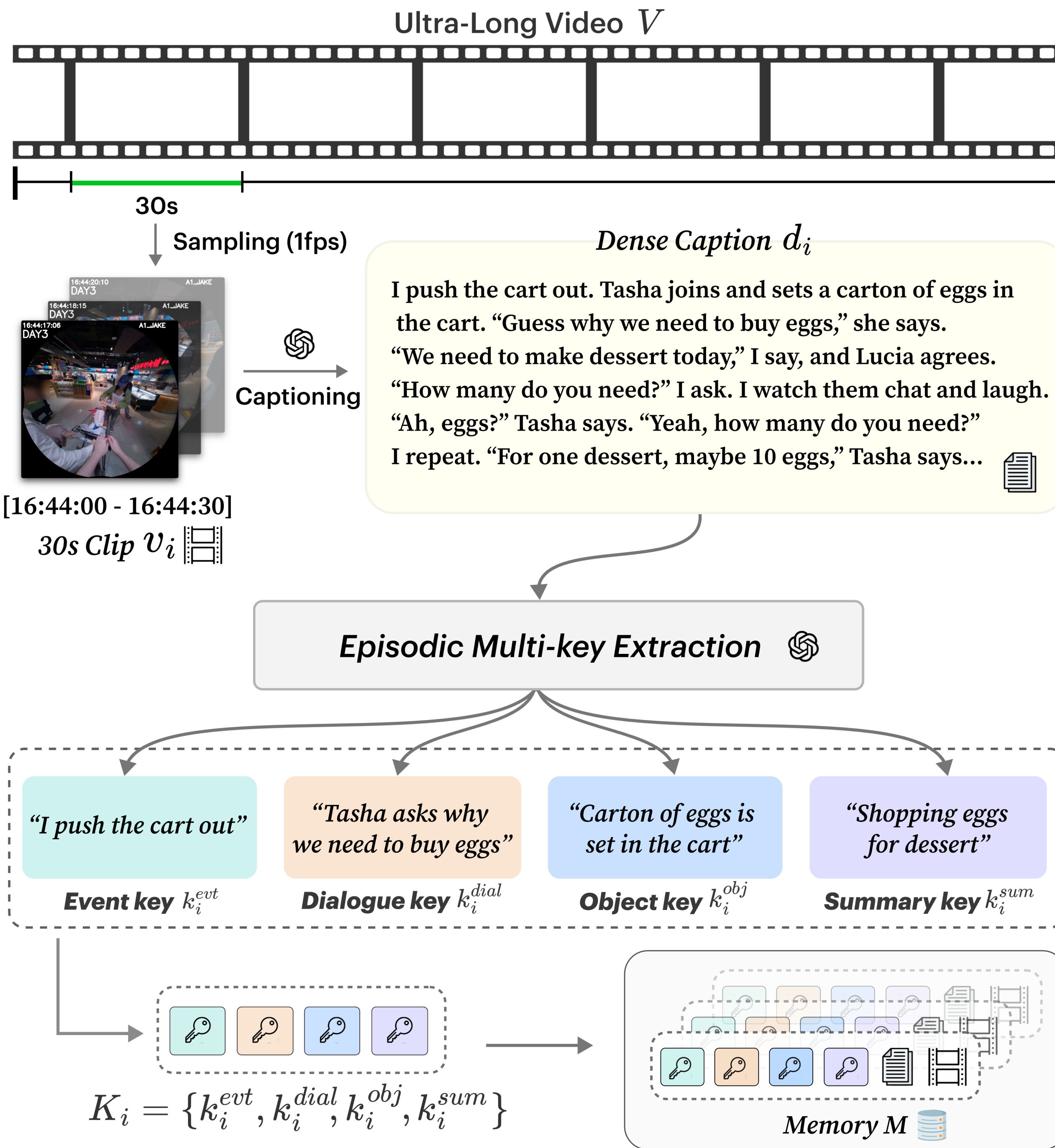
Experiment on EgoLifeQA, NF = Neighbor Filtering

| Setting                           | NF | Acc Avg     | Hit Rate    |
|-----------------------------------|----|-------------|-------------|
| MERIT (Qwen3-VL-8B)               | X  | 48.0        | 0.35        |
| <b>MERIT (Qwen3-VL-8B)</b>        | ✓  | <b>54.2</b> | <b>0.42</b> |
| MERIT (GPT-5)                     | X  | 70.0        | 0.40        |
| <b>MERIT (GPT-5)</b>              | ✓  | <b>71.2</b> | <b>0.56</b> |
| Target Time Oracle (GPT-5)        | X  | 82.4        | 1.00        |
| <b>Target Time Oracle (GPT-5)</b> | ✓  | <b>83.2</b> | <b>1.00</b> |

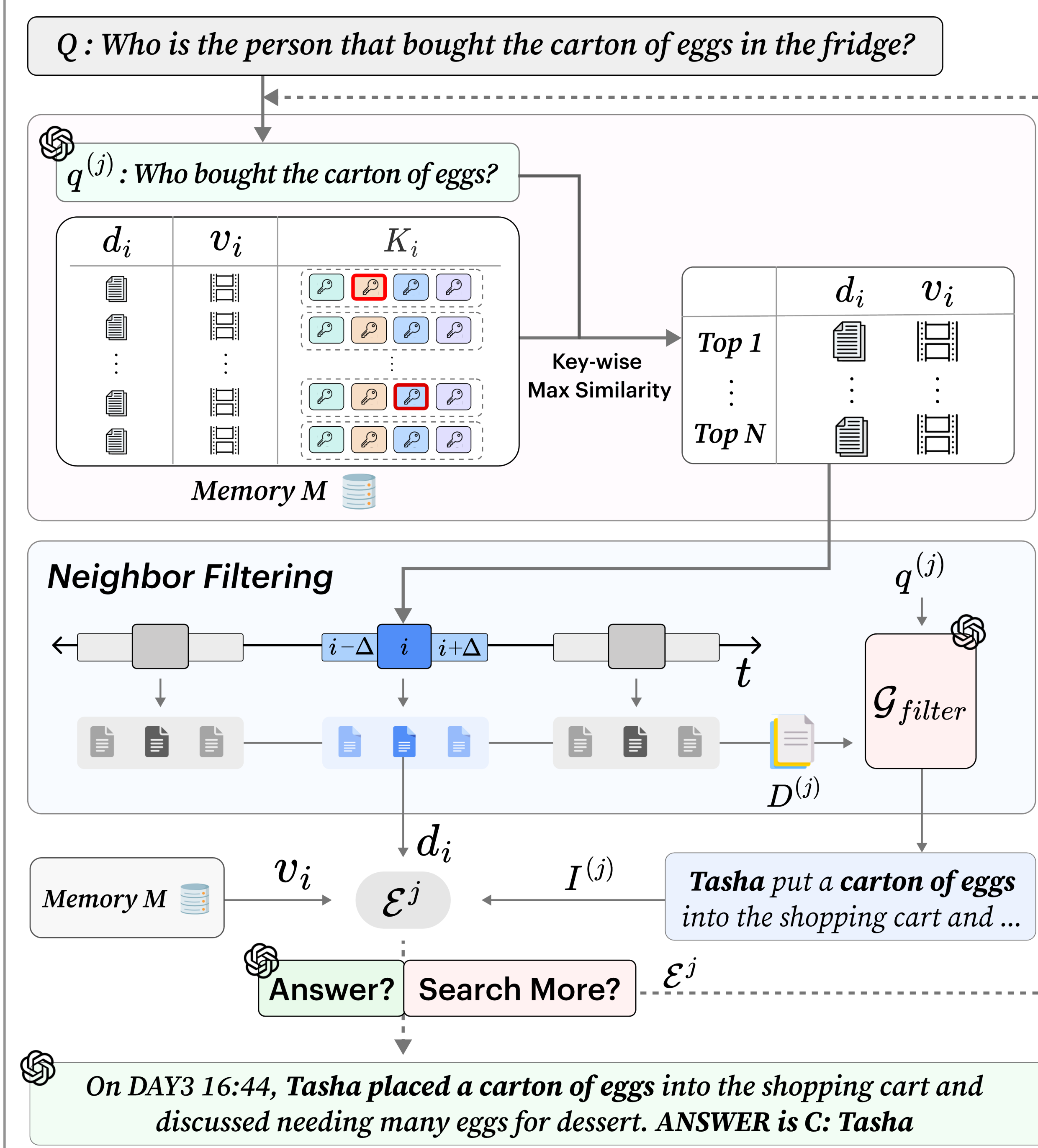
- Expanding temporal context via NF during inference significantly **boosts accuracy, hit rates, and performance in context-heavy QA tasks**.
- Even with perfect Clue Time Evidence (Target Time Oracle), a **single 30s clip is not enough**: 82.4 → 83.2

## MERIT Overall Pipeline

### (a) Stage 1 | Episodic Memory Construction



### (b) Stage 2 | Agentic Retrieval and Answering



## (2) MERIT's Main Component

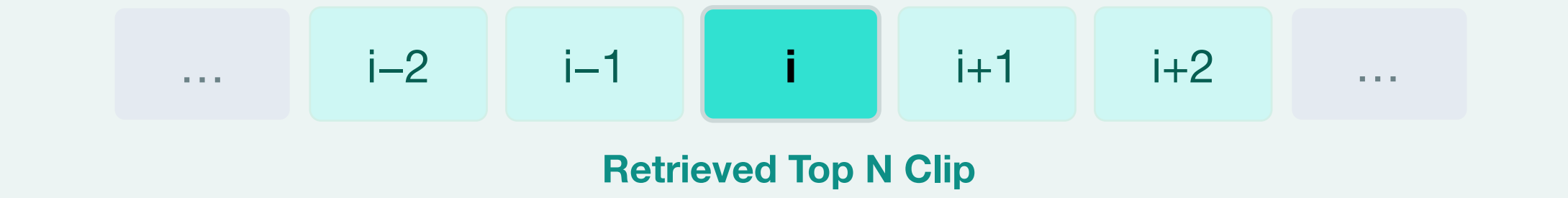
### ① Multi-Key Memory



- Four complementary keys per 30s clip.
- Scored by the best-matching key, ensuring retrieval from any query angle.

**1 Single key 46.4 → 4 Multi-key 54.2** (avg acc., solver Qwen3-VL-8B)

### ② Neighbor Filtering



- Each retrieved anchor pulls in its  $\pm\Delta$  adjacent clips ( $\Delta = 2$ , i.e.  $\pm 1$  min).
- **Only their dense captions** go to the solver — which returns the query relevant evidence, or nothing if the window has none.
- Only one extra LLM call per round.

## (3) Retrieval Accuracy (Hit Rate)

Reproduced under identical 30s captions and captioner

| Method       | Memory Structure          | QA Acc.     | Hit Rate    |
|--------------|---------------------------|-------------|-------------|
| EgoRAG*      | Hierarchy                 | 49.7        | 0.15        |
| WorldMM*     | Multi-scale Graph         | 63.2        | 0.53        |
| <b>MERIT</b> | <b>Flat 30s Multi-key</b> | <b>71.2</b> | <b>0.56</b> |

**Simplest memory, yet the best retrieval !**

## (7) Qualitative Results

